

EJS Template Engine with Express.js (Part 1) –

1. Introduction to EJS

EJS (Embedded JavaScript Templates) is a JavaScript-based template engine used with Express.js to generate dynamic HTML pages.

Key Features of EJS

- Allows embedding JavaScript directly inside HTML
 - Helps create **dynamic, reusable, and clean HTML**
 - Eliminates the need for `<script>` tags for simple logic
 - No need to manually target DOM elements
 - Ideal for server-side rendering with Express.js
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2. Why Use a Template Engine?

Without a template engine:

- HTML files remain static
- JavaScript logic must be added separately

With EJS:

- JavaScript logic can be written directly in HTML
 - Dynamic data can be injected from the server
 - Control structures like loops and conditions can be used in HTML
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3. EJS Syntax and Tags

3.1 Commonly Used EJS Tags

Tag	Purpose
<code><% %></code>	Execute JavaScript code (if, loops, logic)
<code><%= %></code>	Print variable value (HTML escaped – safe)
<code><%- %></code>	Print raw HTML (unsafe, no escaping)
<code><%# %></code>	Comment in EJS
<code><%_ %></code>	Remove whitespace before the tag
<code><% -%></code>	Prevent automatic new line
<code><% %_></code>	Remove whitespace after the tag

Most frequently used tags: `<% %>` and `<%= %>`

4. Printing Values and Loops in EJS

Example: Loop inside HTML

```
<ul>
  <% for(let i = 1; i <= 3; i++) { %>
    <li><%= i %></li>
  <% } %>
</ul>
```

Output:

```
1
2
3
```

Security Note

- `<%= %>` escapes HTML and protects against **XSS attacks**
- `<%- %>` does not escape HTML and is unsafe

5. Installing EJS with Express.js

Step 1: Initialize Node.js Project

```
npm init -y
```

Step 2: Install Required Packages

```
npm install express nodemon ejs
```

6. Project Setup

Folder Structure

```
ejs-project/
|
├─ index.js
├─ package.json
```

```
└─ views/  
  └─ about.ejs
```

7. Configuring Express.js with EJS

Enable ES6 Imports

In `package.json`:

```
"type": "module"
```

index.js Basic Setup

```
import express from "express";  
  
const app = express();  
  
app.set("view engine", "ejs");  
  
app.listen(3000, () => {  
  console.log("Server started on port 3000");  
});
```

By default, Express looks for `.ejs` files inside the **views** folder.

8. Creating Routes and Rendering Views

Static Route Example

```
app.get("/", (req, res) => {  
  res.send("Home Page");  
});
```

Rendering an EJS File

```
app.get("/about", (req, res) => {  
  res.render("about");  
});
```

This automatically loads:

```
views/about.ejs
```

9. Passing Data to EJS Templates

Sending Data from Route

```
app.get("/about", (req, res) => {  
  res.render("profile", {  
    title: "About Page",  
    message: "Welcome to EJS"  
  });  
});
```

Accessing Data in EJS

```
<title><%= title %></title>  
<h1><%= title %></h1>  
<p><%= message %></p>
```

10. Conditional Rendering (if-else)

EJS If Statement

```
<% if(message) { %>  
  <p><%= message %></p>  
<% } else { %>  
  <span>There is no message</span>  
<% } %>
```

- Paragraph is shown only if `message` exists
 - Else block executes if message is empty or null
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11. Using Loops with Arrays

Example: Looping Through an Array

Route

```
app.get("/about", (req, res) => {  
  const items = ["Apple", "Banana", "Cherry"];
```

```
res.render("about", { items });
});
```

EJS Template

```
<ul>
  <% items.forEach(item => { %>
    <li><%= item %></li>
  <% }) %>
</ul>
```

12. Removing Extra Whitespace

To avoid extra new lines:

```
<% items.forEach(item => { -%>
  <li><%= item %></li>
<% }) -%>
```

13. Passing Array of Objects (Users Example)

Data from Route

```
const users = [
  { name: "Aman", age: 25, city: "Delhi" },
  { name: "Rohit", age: 30, city: "Mumbai" },
  { name: "Neha", age: 22, city: "Pune" }
];

res.render("about", { users });
```

14. Displaying Data in a Table

EJS Table Example

```
<table border="1" width="500">
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
```

```
<% users.forEach(user => { %>
  <tr>
    <td><%= user.name %></td>
    <td><%= user.age %></td>
    <td><%= user.city %></td>
  </tr>
<% }) %>
</table>
```

15. Key Takeaways

- EJS allows mixing JavaScript with HTML
 - Express automatically loads EJS files from **views**
 - Data can be passed using **res.render()**
 - Supports:
 - Variables
 - Conditions
 - Loops
 - Arrays
 - Array of objects
 - **<%= %>** is secure and recommended
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